

The lanthanide series—the collective name for elements 57-70—is named after lanthanum, the first element in this series.

Lanthanum

57

La

This lanthanide was discovered by the Swedish chemist Carl Gustaf Mosander in 1839. One of its compounds—lanthanum oxide—is added to the glass used in some camera and telescope lenses to give sharper images. Small amounts of lanthanum are added to some steels to improve impact resistance.

Lanthanides

Once known as the “rare earths,” the lanthanides are, in fact, not rare at all, but they are similar to each other. This made them difficult to separate at first—the name “lanthanide” means “like lanthanum,” and in Greek, *lanthano* means “hidden.”

Bastnäsite

This mineral is a mixture of three metals: yttrium, cerium, and lanthanum, combined with carbon, oxygen, and fluorine. Bastnäsite is used as a source of each of those elements, which are extracted from the ore in a complex process.



Neodymium

60

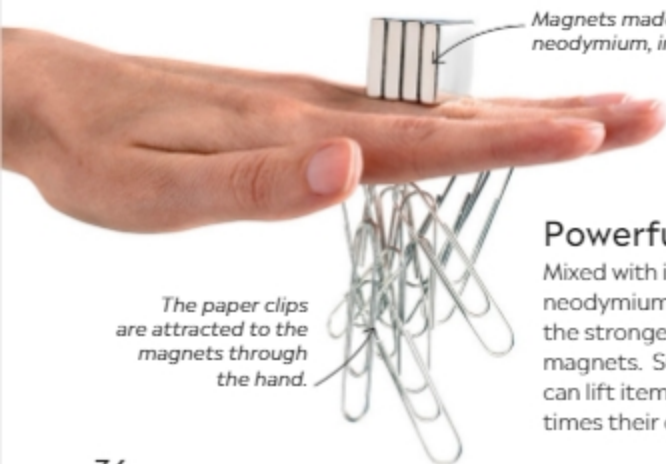
Nd

In 1885, the Austrian chemist Carl Auer von Welsbach extracted this element from didymium, a mixture of neodymium and praseodymium. It's used to make powerful magnets and lasers, including some types of laser pointers.



This silvery-white metal will tarnish quickly in air.

Laboratory sample of pure neodymium



Magnets made from an alloy of neodymium, iron, and boron

Powerful magnets

Mixed with iron and boron, neodymium makes some of the strongest permanent magnets. Such magnets can lift items a thousand times their own weight!

Promethium

61

Pm

Promethium was first discovered in the fission products of uranium in 1945 and was named after Prometheus, the Titan in Greek mythology who stole fire from the gods. At the end of World War II, and with nuclear weapons in mind, this was intended to symbolize the possible dangers of scientific discoveries.

Promethium-rich paint

Paints

Mixed with zinc sulfide, promethium is added to paint to give a green-blue glow. The manufacture of “glow-in-the-dark” watch dials using promethium was once popular, but this application has been largely phased out.

